How the Professor Asks

The grammar of the tutorial, exercise and review questions, the answer template for each verb, the pairs he loves to make you distinguish, and every computation and derivation you must do on paper; with a plan for the day

Fundamentals of Machine Learning

2026-10-30

How to read this document

Every question on the three tutorial sheets, the two exercise sheets and the review list is built from seven verbs. Each verb is graded the same way every time, so each has an answer template. Section 1 gives the seven templates. Section 2 lists the pairs of look-alike objects the professor asks you to distinguish; they account for a third of all his questions. Section 3 is every computation that appears on the sheets, with the number it should give, so you can redo each one until it is automatic. Section 4 is every derivation, with its first line and its one key step. Section 5 is the plan for today. The material itself is in the six Essentials; this document is about the *shape* of the answer.

1 The Seven Verbs and Their Templates

Verb	What is graded	Template
Identify / formulate	you can map an application onto the course's objects	name the object, the representation x, the target Y, the model output, the action, the loss, the learning signal; one line each
Derive	you can go from a definition to the formula	write the definition, substitute, simplify; box the result; say what it means in one sentence
Compute	arithmetic on small numbers	formula first, substitution with units, result, one-sentence reading of the result
Explain why	the mechanism, in the course's vocabulary	claim in one sentence, mechanism in three, consequence in one; use the defined terms
Distinguish A from B	two objects that look alike are different	define each in one line, say what changes one and not the other, give the situation where confusing them hurts
Counterexample / construct	the claim is not always true	give the smallest concrete instance (two points, two distributions, one table) and show the claim failing on it
Audit / redesign	every leak in an experiment, then a protocol	list each mechanism by name (leakage, duplicates, selection, wrong split, wrong threshold); then the protocol step by step

Identify / formulate: "Identify the physical object, x, Y, the model output and the action" (Tutorial 1, Q1(a))

Expert answer

Object: one completed network flow. Representation x: the numerical summary of that flow (duration, packet and byte counts, ports, protocol flags), fixed before the model sees it. Target $Y \in \{\text{benign}, \text{attack}\}$, assigned by the ground-truthing process, so it describes the *episode* the flow belonged to, not the flow's appearance. Model output: an estimate $\hat{p}_\theta(y = 1 \mid x)$, a number in $[0, 1]$. Action: raise an alert or not, chosen from the posterior and the costs, a separate object from the posterior. Learning signal: the labelled pairs (x_i, y_i). One line per object; the marks are for naming all of them and keeping output and action apart.

Explain why: "Why can a classifier using only these features fail to separate them? Are the labels wrong?" (Tutorial 1, Q1(b))

Expert answer

Claim. It cannot, and the labels can still be right. **Mechanism.** The representation maps two physically different events z_A (benign) and z_B (attack) to the same x. A classifier is a function of x alone, so it must give both the same answer; whichever it gives is wrong on one of them. The best possible rule predicts the majority class at that x and still loses the minority share: the *Bayes error*, a property of the representation, not of the learner. **Consequence.** The labels are correct (they describe the episode), the features are insufficient; the fix is a richer x, not relabelling, and no model can beat the Bayes error on this x.

Distinguish: "Why is changing the threshold to reflect costs different from correcting for class imbalance?" (Tutorial 1, Q2(d))

Expert answer

A cost-sensitive threshold takes the posterior as *correct* and changes the *action*: with $C_{FP} = 2$, $C_{FN} = 8$ you alert when $p > 0.2$ because the expected cost of silence, $8p$, exceeds the expected cost of alerting, $2(1 - p)$. Correcting for imbalance changes the *estimate*: reweighting or resampling alters the training distribution so that $\hat{p}$ is no longer the deployment posterior, which is useful when the minority class is too small to estimate well, or when the deployment prior differs from the training prior (prior shift, Tutorial 1 Q6(e)). Confusing them hurts when both are applied: a resampled model's $\hat{p}$ is already inflated, and a low threshold on top of it floods the analysts with alerts.

Counterexample: "Does hard voting always lose to soft voting?" (Tutorial 2, Q1(b))

Expert answer

No. Flow 1: $p_A = p_B = 0.49$, $p_C = 0.95$, $y = 1$. Hard voting: two votes benign, predicts 0, wrong. Soft voting: mean 0.643, predicts 1, right; it used the confidence that the two near-ties discarded. Now let C be over-confident and wrong: flow 3, $p = (0.40, 0.40, 0.90)$, $y = 0$. Hard: benign, right. Soft: mean 0.567, attack, wrong. One over-confident member drags the average; majority voting caps any single member at one vote. So soft voting wins when probabilities are calibrated, hard voting is more robust to a badly calibrated member; neither always wins.

Audit: "Identify every mechanism that makes the reported accuracy too optimistic" (Tutorial 1, Q4(a); Exercises ch. 2, Q10)

Expert answer

Name each one. (1) **Preprocessing leakage**: the scaler was fitted on all records, so test statistics shaped the training inputs. (2) **Duplicate leakage**: near-identical flows of one episode land on both sides of a row-wise split, so the test set contains the training set. (3) **Selection on the test set**: the best of 100 (or 500) configurations on the test set is a validation score, and its optimism is the expected maximum of 100 noisy draws (Q5(d) gives the two-configuration case: $R - \sigma/\sqrt{\pi}$). (4) **Wrong split for the goal**: random rows cannot speak for later traffic (temporal) or new episodes (grouped). (5) **Wrong metric and threshold**: accuracy on 1% attacks is 0.99 for the all-benign rule, and a 0.5 threshold ignores a 100:1 cost ratio. The protocol: split by episode (and by time when drift is expected), fit scaler, imputer and feature selection inside each training fold, select on validation, report the test set once, with precision, recall, F_1 and the cost-weighted error at the chosen threshold.

2 The Pairs He Makes You Distinguish

One line each: what each is, and what moves one but not the other.

- **Posterior versus decision.** $\hat{p}(y{=}1 \mid x)$ is a number; the alert is an action chosen from that number and the costs. Costs move the decision, not the posterior.
- **Target leakage versus shortcut.** Leakage: a feature that is a consequence of the label or unavailable at decision time (the incident ticket id). Shortcut: a legitimate feature that is correlated with the label in the collected data but not in deployment (time of day, snow in wolf photos). Leakage is removed by construction; a shortcut is detected by data that breaks the correlation.
- **Training loss versus decision loss.** Cross-entropy trains the probability estimate; zero-one or cost-weighted loss judges the decision. They can rank two models differently (Exercises ch. 2, Q4) and that does not make cross-entropy wrong: it estimates the quantity the decision needs.
- **Threshold for costs versus correction for imbalance versus prior shift.** Threshold: same $\hat{p}$, different action. Imbalance correction: changes how $\hat{p}$ is estimated when the minority is tiny. Prior shift: re-weights a calibrated $\hat{p}$ by π_d/π_t because the deployment base rate differs.
- **Empirical risk versus population risk versus generalization gap.** Training average of the loss (observable), expectation under the true distribution (not), their difference (not). Only the first is seen during training.
- **Validation versus test.** Validation selects (and is therefore optimistic once used for selection); test is touched once, after all choices.
- **Underfitting versus overfitting.** High training and validation error with a small gap; low training error with a large gap. The cure for one worsens the other.
- **Bias versus variance at a fixed x.** Bias: how far the *average* prediction over repeated training samples is from the truth. Variance: how much the prediction *spreads* over those samples. Noise: the part no predictor removes.
- **Parameter versus hyperparameter.** Fitted on training data by the learning algorithm versus chosen on validation data by the designer; the same quantity (polynomial degree) can be either, depending on the procedure (Exercises ch. 2, Q8).
- **Random versus grouped versus temporal split.** Rows exchangeable; rows clustered (episodes, patients); the future differs from the past.
- **Accuracy versus precision versus recall.** Fraction right overall; fraction of alerts that are attacks; fraction of attacks that were alerted. Imbalance makes the first meaningless.

- **Discriminative versus generative.** Models $p(y \mid x)$ directly (logistic regression) versus models $p(x \mid y)\,p(y)$ and applies Bayes (LDA, naive Bayes). The second needs more assumptions, uses data more efficiently when they hold, can flag unfamiliar x, handles missing features; the first is robust to a wrong $p(x \mid y)$.
- **LDA versus QDA versus naive Bayes.** Shared covariance gives a linear boundary; separate covariances a quadratic one; conditional independence gives a product of one-dimensional likelihoods, cheap and often well-ranked but over-confident when features repeat evidence.
- **Hard versus soft voting.** Counts labels versus averages probabilities; the second uses confidence, the first is robust to one miscalibrated member.
- **Bagging versus boosting.** Independent fits on bootstrap samples, averaged, to cut variance; sequential fits to residuals, summed with shrinkage, to cut bias. Random forests add feature subsampling to decorrelate the trees.
- **MDI versus permutation importance.** Impurity credit summed over the training splits (biased toward high-cardinality features, computed on training data) versus the validation error increase when one feature is shuffled (measures reliance, splits credit between correlated features). Neither proves causation.

3 The Computations, With Their Answers

Redo each until the number comes out without looking. Costs, counts and probabilities are from the sheets.

1. **Cost threshold** (T1 Q2(a)). Risk of alerting $= C_{FP}(1-p)$, of silence $= C_{FN}\,p$. Alert when $p > C_{FP}/(C_{FP}+C_{FN}) = 2/10 = 0.2$. Flows 0.08 / 0.35 / 0.85: benign, attack, attack.
2. **Three regions with review** (T1 Q2(c), Ex. 2 Q3). Review costs 0.9. Benign when $8p < 0.9$ and $8p < 2(1-p)$: $p < 0.1125$. Attack when $2(1-p) < 0.9$: $p > 0.55$. Review between. With costs 1 / 9 / 0.6: benign $p < 0.067$, attack $p > 0.4$, defer between. One threshold is no longer enough because the middle action has a flat risk that undercuts both sloped ones.
3. **Nearest neighbour and scaling** (T1 Q3, Ex. 2 Q5). Raw: $d_A = \sqrt{10^2+90^2} = 90.6$, $d_B = \sqrt{350^2+2^2} = 350.0$: A. Scaled by 100 and 20: $x = (0.5, 0.5)$, $A = (0.6, 5)$, $B = (4, 0.6)$; $d_A = 4.50$, $d_B = 3.50$: B. Weighted $w(z_1 - z_1')^2 + (z_2 - z_2')^2$: equal when $0.01w + 20.25 = 12.25w + 0.01$, $w = 1.65$; below it B, above it A. The people and flows did not change; the geometry did, which is why w is a hyperparameter chosen on validation data.
4. **Confusion metrics** (T1 Q4(c)). TP 70, FP 180, FN 30, TN 9,720: accuracy 0.979, precision $70/250 = 0.28$, recall $70/100 = 0.70$, $F_1 = 2 \cdot 0.28 \cdot 0.70/0.98 = 0.40$. All-benign rule: accuracy 0.99, recall 0.
5. **Expected minimum of two noisy validations** (T1 Q5(d)). $\mathbb{E}[\min] = R + \mathbb{E}[\varepsilon_1 + \varepsilon_2]/2 - \mathbb{E}|\varepsilon_1 - \varepsilon_2|/2 = R - \sigma/\sqrt{\pi} = 0.20 - 0.0113 = 0.189$. Selecting the smaller of two equal-quality numbers reports less than the truth; with 100 configurations the optimism is larger.
6. **Sigmoid and cross-entropy** (T1 Q6(a)). $\sigma(1.2) = 1/(1+e^{-1.2}) = 0.769$; $\ell = -\ln 0.769 = 0.263$. Threshold with $C_{FP} = 4$, $C_{FN} = 16$: 0.2, attack. Boundary of $s = -3 + 2x_1 + x_2$ at 0.5: $2x_1 + x_2 = 3$; at $p = 0.2$: $s = \ln(0.2/0.8) = -1.386$, so $2x_1 + x_2 = 1.614$, the line moves toward the benign side and the attack region grows.
7. **Prior shift** (T1 Q6(e)). $\pi_t = 0.20$, $\pi_d = 0.05$, $r_t = 0.60$: numerator $0.25 \times 0.6 = 0.15$; denominator $0.15 + (0.95/0.80) \times 0.4 = 0.625$; $r_d = 0.24$.
8. **LDA with a shared covariance** (T1 Q7 I). $\rho = 0.5$: $\Sigma^{-1} = \frac{1}{0.75}\begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix}$. $s_0 = \ln 0.75$; $s_1 = \frac{4}{3}(x_1 + x_2) - \frac{8}{3} + \ln 0.25$. Boundary $x_1 + x_2 = 2 + \frac{3}{4}\ln 3 = 2.82$. At $(1.25, 1.25)$: sum 2.5,

class 0; $p_1 = \sigma(s_1 - s_0) = \sigma(-0.432) = 0.39$. Equal priors: boundary $x_1 + x_2 = 2$, class 1, $p_1 = \sigma(0.667) = 0.66$. The prior shifts the line along the direction $\Sigma^{-1}(\mu_1 - \mu_0)$; the class models did not move.

9. **Naive Bayes** (T1 Q7 II). Attack: $0.6 \times 0.7 \times 0.1 = 0.042$; benign: $0.05 \times 0.2 \times 0.9 = 0.009$; posterior attack $0.042/0.051 = 0.82$.

10. **Best point prediction under a loss** (Ex. 2 Q2). Exponential with mean 4: squared loss gives the mean 4; absolute loss the median $4 \ln 2 = 2.77$; the asymmetric loss (4 per unit under, 1 per unit over) the 0.8-quantile, $-4 \ln 0.2 = 6.44$. The loss, not the data, decides.

11. **Zero-one versus cross-entropy** (Ex. 2 Q4). A: 0 errors, CE $= -\ln 0.6 = 0.51$. B: 1 error, CE $= (3 \times 0.010 + 1.609)/4 = 0.41$. B is better calibrated and worse at 0.5.

12. **Neighbours in high dimension** (Ex. 2 Q6). $M_d \sim \text{Binomial}(n, r^d)$, $\mathbb{E}M_d = nr^d$, $\mathbb{P}(M_d \geq 1) = 1 - (1 - r^d)^n$. $n = 1000$, $r = 0.1$: $d = 3$ gives 0.63, $d = 4$ gives 0.095: first failure at $d = 4$. For 0.95: $r_d = (1 - 0.05^{1/n})^{1/d} \approx (3/n)^{1/d}$, which climbs toward 1 as d grows: every point becomes a far neighbour.

13. **Variance of an average** (Ch. 5). M predictors, variance σ^2, pairwise correlation ρ: Var $= \rho\sigma^2 + (1 - \rho)\sigma^2/M$. Averaging kills only the uncorrelated part.

14. **Bootstrap and out-of-bag** (Ch. 5). $\mathbb{P}$(a row is left out) $= (1 - 1/N)^N \to e^{-1} = 0.368$; each tree has about a third of the rows as a free validation set.

15. **Random forest feature availability** (T2 Q2(a)). $m_{try} = 5$ of $d = 25$: available at a split with probability 0.2; unavailable at all $M = 10$ roots with $0.8^{10} = 0.107$.

16. **MDI credit and permutation importance** (T2 Q3). Root: $1 \times (0.50 - 0.45) = 0.05$; deeper node: $0.30 \times (0.48 - 0.08) = 0.12$, more credit from a bigger drop on fewer rows. Permutation: $0.20 - 0.12 = 0.08$ for x_a, $0.14 - 0.12 = 0.02$ for x_b.

17. **One boosting step** (T2 Q4). $y = (0, 2, 4, 6)$, $F_0 = 3$, residuals $(-3, -1, 1, 3)$, leaves $x \leq 2.5$: -2, $x > 2.5$: $+2$. $\eta = 0.5$: $F_1 = (2, 2, 4, 4)$, residuals $(-2, 0, 0, 2)$, squared error 20 before, 8 after.

18. **Newton leaf value** (T2 Q5). $g_i = p_i - y_i$, $h_i = p_i(1 - p_i)$, leaf value $-\sum g / \sum h = \sum(y_i - p_i)/\sum p_i(1 - p_i)$.

4 The Derivations, With Their First Line and Key Step

1. **Cost threshold.** Start: conditional risk $R(a \mid x) = \sum_y \ell(y, a)\, p(y \mid x)$. Key step: set $C_{FP}(1 - p) = C_{FN}\, p$.

2. **Mean, median, quantile.** Start: $a^\star = \arg\min_a \mathbb{E}[\ell(Y, a) \mid x]$. Key step: differentiate under the expectation; squared loss gives $a = \mathbb{E}[Y \mid x]$, absolute loss gives $\mathbb{P}(Y < a) = 1/2$, the asymmetric loss with weights c_u (under) and c_o (over) gives $\mathbb{P}(Y \leq a) = c_u/(c_u + c_o)$.

3. **Cross-entropy is maximum likelihood** (T1 Q6(b)). Start: i.i.d. conditional likelihood $\prod_i \hat{p}(y_i \mid x_i)$. Key step: take $-\frac{1}{n}\log$; the product becomes the average of $-\log \hat{p}(y_i \mid x_i)$, which for a binary label is $-y \log \hat{p} - (1 - y)\log(1 - \hat{p})$.

4. **Logistic gradient.** Start: $\ell = -y \log \sigma(s) - (1 - y)\log(1 - \sigma(s))$, $s = \theta^\top x$. Key step: $\sigma' = \sigma(1 - \sigma)$, so $\partial\ell/\partial s = \sigma(s) - y$ and $\nabla_\theta \ell = (\sigma(s) - y)\, x$. The same first line gives $\partial^2\ell/\partial s^2 = \sigma(1 - \sigma)$, the h_i of boosting (T2 Q5).

5. **LDA score** (T1 Q7(a)). Start: $\log p(y = c \mid x) = \log \pi_c - \frac{1}{2}(x - \mu_c)^\top \Sigma^{-1}(x - \mu_c) + \text{const}$. Key step: expand the quadratic; $x^\top \Sigma^{-1} x$ is common to all classes and drops, leaving $x^\top \Sigma^{-1}\mu_c - \frac{1}{2}\mu_c^\top \Sigma^{-1}\mu_c + \log \pi_c$, linear in x; with class-specific Σ_c the quadratic term stays (QDA).

6. **Bias–variance decomposition.** Start: $\mathbb{E}_{\mathcal{D},\varepsilon}[(Y - \hat{f}_{\mathcal{D}}(x))^2]$ with $Y = f(x) + \varepsilon$. Key step: add and subtract $\mathbb{E}_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(x)]$; the cross terms vanish because ε has mean zero and is independent of $\mathcal{D}$; the result is $\sigma_\varepsilon^2 + (\mathbb{E}\hat{f} - f)^2 + \mathrm{Var}(\hat{f})$.
7. **Expected minimum of two noisy scores** (T1 Q5(d)). Start: $\min(a, b) = (a + b - |a - b|)/2$. Key step: $\varepsilon_1 - \varepsilon_2 \sim N(0, 2\sigma^2)$, whose absolute value has mean $2\sigma/\sqrt{\pi}$.
8. **Variance of an average of correlated predictors.** Start: $\mathrm{Var}(\frac{1}{M}\sum_m \hat{f}_m) = \frac{1}{M^2}\sum_{m,m'} \mathrm{Cov}$. Key step: M diagonal terms σ^2 and $M(M-1)$ off-diagonal terms $\rho\sigma^2$.
9. **Why the minimum training error cannot increase in nested families** (T1 Q5(a), Review Q5). Start: $\mathcal{H}_1 \subseteq \mathcal{H}_2$. Key step: the minimiser over $\mathcal{H}_1$ is a candidate in $\mathcal{H}_2$, so the minimum over $\mathcal{H}_2$ is at most that value.
10. **Newton leaf value** (T2 Q5). Start: second-order expansion of the loss in the leaf's added value w: $\sum_{i\in\text{leaf}}(g_i w + \frac{1}{2}h_i w^2)$. Key step: set the derivative in w to zero.

5 Where the Questions Come From

The sheets are professor-written, not copied from a book: none of their titles, set-ups or numbers appears anywhere in print, the decks carry no bibliography, and the exercises reuse the lectures' own examples (port 22, the time-of-day shortcut, the prior-shift formula). What they do reuse is the *canonical treatment* of each topic, so when an answer in the Essentials is not enough, the original is here (all free from the authors):

- **Lindholm, Wahlström, Lindsten, Schön, *Machine Learning: A First Course for Engineers and Scientists*** (smlbook.org): the closest book to this course in level and order. Ch. 2 (k-NN, scaling), 3–4 (logistic regression, evaluation, bias–variance, imbalance), 5 (regularization, early stopping), 7 (bagging, forests, boosting).
- **Murphy, *Probabilistic Machine Learning: An Introduction*** (probml.github.io): §5.1 Bayes decision theory with the reject option and cost thresholds (Tutorial 1 Q2, Exercises ch. 2 Q3), §4.5.5 cross-validation and leakage, §9.2–9.3 LDA and naive Bayes, §10.2–10.3 logistic regression as maximum likelihood, §18.3–18.4 forests and boosting, §19.5 prior (label) shift (Tutorial 1 Q6(e), from Saerens et al. 2002).
- **Hastie, Tibshirani, Friedman, *The Elements of Statistical Learning*** (free at hastie.su.domains): §2.4–2.5 decision theory and the curse of dimensionality (Exercises ch. 2 Q2, Q6), §7.3 and §7.10 bias–variance and "the wrong and right way to do cross-validation" (Tutorial 1 Q4–Q5), §10.10–10.12 gradient boosting with shrinkage (Tutorial 2 Q4), ch. 15 random forests, out-of-bag and variable importance (Tutorial 2 Q2–Q3).
- **Bishop, *Pattern Recognition and Machine Learning***: §1.5 decision theory, reject option, mean versus median under the loss; §4.2–4.3 generative and discriminative classifiers.
- **Two papers behind Tutorial 2**: Chen and Guestrin 2016, *XGBoost*, §2.2 for $g = p - y$, $h = p(1-p)$ and the Newton leaf value (Q5–Q6); Louppe 2014 for the MDI credit formula, which is scikit-learn's `feature_importances_`. The wolves-and-snow story of Exercises ch. 1 Q4 is Ribeiro et al. 2016 (LIME), §6.4.

Problem by problem

Verified by reading the sources. "Adapted" means the same structure with new numbers or a new story; "topic" means only the subject matches.

Problem	Closest published exercise	Match
T1 Q2	Edinburgh PMR decision-making sheet Ex 3 (threshold $c_{FP}/(c_{FP}+c_{FN})$); Murphy PML1 Ex 5.1 reject option; Bishop Ex 1.24	adapted
T1 Q3, Ex2 Q5	Uppsala SML session 3, Ex 3.1 (k-NN by hand with a distance table)	adapted
T1 Q5(a)–(c)	Uppsala SML session 5, Ex 5.4–5.5 (nested families, train versus hold-out error)	adapted; (d) is original
T1 Q6	Uppsala SML 3.2 (logistic model: probability, 0.5 boundary); Bishop Ex 4.13 gradient; prior shift from Saerens et al. 2002	adapted
T1 Q7 I	ISLR Ex 4.2 (Bayes classifier as discriminant); CS229 PS1 GDA (shared Σ gives a linear boundary)	adapted
T1 Q7 II	Uppsala SML 3.4 (Bayes with priors by hand); CS229 PS2 naive Bayes with smoothing	adapted
T1 Q1, Q4, Q8	LFD Ex 1.1, Mitchell Ex 1.2; ESL §7.10.2; none	topic / original
T2 Q1	ISLR Ex 8.5 (ten estimates of $P(\text{Red} \mid X)$: majority vote versus average probability)	adapted, near-copy of structure
T2 Q2	ISLR Ex 5.2 and Uppsala 7.2 (exclusion probability chain $(1-p)^M$); ISLR §8.2.2 on m_{try}	adapted
T2 Q4	ISLR Ex 8.2 and Algorithm 8.2 (stumps, residuals, shrinkage)	adapted
T2 Q5–Q6	Chen and Guestrin 2016 §2.2, XGBoost docs "Introduction to Boosted Trees"	derivation near-copy
T2 Q3, Q7, Q8, Q9	scikit-learn MDI formula and Louppe 2014; Uppsala 5.4 and 7.2(d); none	topic / original
Ex2 Q2	Edinburgh PMR Ex 1 (mean, median); Murphy Ex 5.2 newsvendor (asymmetric loss gives a quantile)	adapted
Ex2 Q3	Murphy PML1 Ex 5.1 (reject option, regions, rising reject cost)	adapted, near-copy of structure
Ex2 Q6	ESL Ex 2.3 (nearest neighbour in the unit ball); Uppsala 3.7	adapted
Ex2 Q1, Q7, Q9, Q10	Bishop §1.5.1; LFD ch. 1; Shalev-Shwartz ch. 2; ESL §7.10.2	topic
Ex2 Q4, Q8	none	original
Ex1 Q4, Q5, Q6, Q9	LIME paper §6.4 (wolves and snow); LFD Ex 1.6 and 1.1; Mitchell Ex 1.2	adapted
Ex1 others	thought experiments with no published exercise	topic / original

The Uppsala sheets (uu-sml.github.io/course-sml-public, sessions 3, 5, 7, with solutions) are the closest thing to a second set of his problems with different numbers.

6 The Day

Nine working hours, in blocks of fifty minutes

1. **Hours 1–2, chapters 1 and 2.** Read the two Essentials. Then redo computations 1, 2, 3, 10, 11 above on paper. Then Exercises ch. 2 Q3 and Q4 from the sheet with the solutions closed.
2. **Hour 3, chapter 3.** Read the Essential. Answer the 20 review questions aloud, one sentence each; mark the ones you could not, reread only those parts. Redo computations 4 and 5.
3. **Hour 4, chapter 4.** Read the Essential. Redo computations 6, 7, 8, 9; derive the LDA score once by hand (derivation 5). Then Tutorial 1 Q6 and Q7 from the sheet, solutions closed.

4. **Hour 5, chapter 5.** Read the Essential. Redo computations 13 to 17; derivation 4 and 10. Then Tutorial 2 Q4 and Q5 from the sheet, solutions closed.
5. **Hour 6, the verbs.** Section 1 and 2 of this document. For ten of the distinguish pairs, write the two-line answer from memory.
6. **Hour 7, mock.** Tutorial 1 as an exam: two hours' worth in one, skipping what you did in hours 2 and 4. Grade yourself with the solutions PDF.
7. **Hour 8, the gaps.** Whatever the mock exposed: that chapter's Essential section plus its drill box.
8. **Hour 9, close.** The five summary boxes, the sixteen pairs, the cheat sheet. Stop.

Every hour ends with five minutes of writing, from memory, the three things you just learned. Nothing after midnight; a rested reader of these pages beats a tired reader of the full chapters.